

Deepak Mallampati

APPLIED AI / LLM ENGINEER

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Kent, OH, USA - F-1 OPT (open to relocation)

PROFESSIONAL SUMMARY

Applied AI / LLM Engineer with 4+ years building production LLM applications - vector-database retrieval, single- and multi-agent systems, and LLM-as-a-judge evaluation pipelines. Ship reliable GenAI features end-to-end: dense + cross-encoder retrieval reducing hallucination by **40%** at **92% precision**, conductor-subagent agent orchestration cutting token cost **42%**, and RAGAS/BERTScore eval frameworks that catch regressions before release. Hands-on with OpenAI, Anthropic, and Gemini APIs and HuggingFace transformers, with Python-first engineering discipline and Kubernetes deployment.

CORE COMPETENCIES & SKILLS

LLM Applications • Vector Databases (FAISS/Pinecone/Weaviate) • Single & Multi-Agent Systems • LLM-as-a-Judge Evaluation • RAG / Retrieval • OpenAI / Anthropic / Gemini APIs • HuggingFace Transformers • Data Pipelines • Python • Kubernetes

Generative AI & LLMs: GPT-4o/GPT-5, Claude Sonnet, Gemini, LLaMA | LangChain/LangGraph | RAG | Single & Multi-Agent Systems | Prompt Engineering | Fine-tuning (LoRA/QLoRA/PEFT)

Vector Search & Retrieval: FAISS, Pinecone, Weaviate | Semantic & Hybrid Search | Cross-Encoder Re-ranking | Document Chunking | Semantic Caching

LLM Evaluation: LLM-as-a-Judge | RAGAS | BERTScore | Eval datasets | Drift Detection | A/B Testing | Latency/Accuracy Dashboards

ML & NLP: PyTorch, TensorFlow, Keras | HuggingFace Transformers | NER, Text Classification, Summarization | Model Optimization

MLOps & Cloud: AWS (Bedrock, SageMaker, Lambda), Azure OpenAI, GCP Vertex AI | Docker, Kubernetes | Terraform | CI/CD | MLflow, W&B

Data Engineering: PostgreSQL, MongoDB, Redis | Spark, Databricks | Kafka | Pandas, NumPy | Data Lineage & Governance

Languages & Tools: Python, JavaScript/TypeScript, Java, SQL | FastAPI, React | Git, Jira, Confluence | Agile/Scrum

PROFESSIONAL EXPERIENCE

Progress Solutions

Jul 2025 - Present

AI Engineer

USA

- Architected production **multi-agent LLM systems** on a conductor-subagent orchestration framework, decomposing objectives into context-scoped subagents that execute autonomously - reducing token consumption **42%** while sustaining task accuracy across multi-step workflows.
- Engineered **RAG pipelines** over large healthcare corpora with dense vector retrieval + cross-encoder re-ranking, lifting answer relevance and materially reducing hallucination on domain queries.
- Established an **LLM evaluation framework** (LLM-as-a-judge, RAGAS, BERTScore, custom metrics) with eval datasets and latency/accuracy dashboards to benchmark iterations pre-deployment and surface regressions before release.
- Built fault-tolerant data + automation pipelines (scheduled batch orchestration, exponential-backoff retry), reducing failed production runs **60%**.

Vanguard

Jan 2024 - Jan 2025

AI Engineer Intern

USA

- Developed **AI agents and backend services** for Vanguard's internal AI platform across **25+ agents** and workflows, integrated with production data pipelines and observability tooling.
- Evaluated **GPT-4o, Claude Sonnet, and Gemini** via prompt optimization, improving task success **27%** and informing platform model-selection.
- Built **12 APIs and microservices**, reducing p95 latency 1.5s → **800ms** and supporting **100,000+ requests daily** across 20+ teams.

Kent State University

Oct 2023 - Jan 2024

ML & Gen AI Research Assistant

USA

- Fine-tuned **Qwen3 via 4-bit QLoRA** on an H100 with a curated instruction dataset for controllable long-form generation.
- Built a privacy-preserving ML pipeline (k-anonymity, l-diversity, differential privacy), cutting re-identification risk 3.38% → 0.32% while retaining **99%** of baseline predictive performance.

Emerson

Jun 2022 - Apr 2023

ML Trainee

India

- Fine-tuned **BERT and RoBERTa** (HuggingFace transformers) for entity extraction and classification, achieving **89% F1** on custom NER.
- Trained regression/classification models for equipment-failure prediction and anomaly detection over operational sensor data.

PROJECTS

MangaLens

Shipped Chrome extension (live on the Chrome Web Store) delivering real-time AI translation across 29+ sites - an end-to-end LLM PoC taken to production, with back-end services and an inline front-end UX.

TypeScript/JavaScript · LLM APIs · Chrome Web Store

career-ops

Autonomous AI job-search pipeline that scans listings, scores fit with agentic workflows, and generates tailored applications overnight unsupervised.

Python · Multi-Agent Orchestration · LLM Evals

Story Director LLM

Fine-tuned LLM that generates structured short-form video narratives from a single prompt, paired with an automated rendering pipeline.

QLoRA · HuggingFace · Python

EDUCATION

M.S. in Computer Science - Kent State University, OH

2025

GPA: 3.70 / 4.0